

# Do Epistemic Verbs Improve Prediction? Evidence from Repeated Rock Paper Scissors

Abdul Ahad, *Pragmatic Processing in LLMs, Universität des Saarlandes*

## 1. Motivation

Riemer et al. (2025) argues that theory of mind (ToM) benchmarks for large language models measure the wrong thing. They split ToM into two parts. Literal ToM is whether the model correctly predicts what the other player will do. Functional ToM is whether the model then uses that prediction to act well. Their key example is repeated Rock Paper Scissors: against an opponent that always plays the same move, models predict that move well, but still play all three moves about equally often, so they never exploit it. There is a gap between knowing and acting.

I concluded that the gap isn't that, instead it's the model's ability to find the correct pattern (Prediction), once the correct prediction is elicited, the application of it is near perfect. Furthermore, from my latest experiments I concluded that the prediction ability is hampered by the model's ability of cross-stream reasoning (looking at its own moves w.r.t the opponents move), when this gap is bridged through pointing the prediction approaches the ceiling as well.

To this end, an extension of the experiment was to check if different Verbs change the accuracy and how the model reasons built from the Hypothesis: that different verbs like -predict, anticipate, guess, forecast, infer - that are semantically very close sit at different points on a scale of epistemic force: guess might signal low confidence and little evidence, while predict might imply a more articulate, model based commitment. My hunch is that these verbs do not affect a language model uniformly: some of them may "switch on" the model's prediction behavior better than others, even though the literal task is more or less identical. This could reflect properties of the training data -for instance, guess may co-occur with low-confidence contexts, while predict is more articulate, or one verb may simply appear in patterns that trigger next-word prediction more strongly than another.

## 2. Research question

Does the epistemic verb used to ask for a prediction (*predict*, *infer*, *guess*, ...) change how accurately the model reasons about its opponent's next move? And if the verb changes how the model reasons, does it also show up in accuracy?

## 3. Setup

**Model and game.** Qwen3.5-9B (base and instruct) was used with vLLM. Rock Paper Scissors renamed as J, F and M (so the model cannot rely on memorised Rock Paper Scissors strategies) was played. Each game was a 100 rounds, and the model's own moves were scripted from a fixed seed and the opponent follows a *fixed rule*, so the game history is known in advance and is exactly the same for every prompt. Any difference between two prompts therefore comes from the verb alone.

**Opponents.** Opponent was built on 5 strategies with varying difficulty levels. *Counter* plays what beats the models move in the current round, *Mirror* copies the model's last move, *Tft* plays the move that beats the model's last move, *Cycle* repeats a fixed sequence and ignores the model, and *Random* has nothing to predict and serves as a control. Mirror, counter and tft can only be solved by looking at the model's own past moves w.r.t opponents last moves (cross stream reasoning), which is the hard case for the model.

**Study 1: verb comparison.** Nine epistemic verbs in three categories (self-assigned relative prediction force) were taken, high (predict, forecast, infer), mid (expect, anticipate, estimate) and low (guess, suppose, imagine), only the verb was varied and was prompted in four ways:

"What do you {verb} the other player will play in this round?", "{Verb} the other player's move in this round.", "{Verb} the other player's move in round {N}.", and "Your task is to {verb} the other player's move in this round."

Accuracy was read from free answers and from the model's probability for each move. Some prompts also added a hint sentence, "The opponent's move depends on what you played in the previous round", which is true for mirror and tft and false for cycle and random (induced cross-stream).

**Study 2: predict against infer.** In Study 1's reasoning texts, *infer* was the verb that did cross-stream with the right round lag about twice as often as any other verb. So this was tested directly by changing one word in the task line of an earlier experiment, "Your task is to predict / infer the opponent's next move from the history", at six different hint levels (Table 3).

## 4. Results

In the larger main run (126 opponent by hint cells), no verb is reliably best: *predict* ranked top in 6 cells and bottom in 6.

**Table 1.** how often each verb came first minus how often it came last, over 126 cells.

| verb       | first | last | net |
|------------|-------|------|-----|
| anticipate | 21    | 12   | +9  |
| estimate   | 17    | 15   | +2  |
| imagine    | 16    | 17   | -1  |
| say        | 16    | 14   | +2  |
| expect     | 11    | 11   | 0   |
| guess      | 10    | 15   | -5  |
| infer      | 10    | 14   | -4  |
| suppose    | 10    | 6    | +4  |
| forecast   | 9     | 16   | -7  |
| predict    | 6     | 6    | 0   |

126 cells. Under chance each verb is first 12.6 times.

What the verb does change is the way the model reasons. High verbs reason in terms of rules (*pattern*, *cycle*), mid and low verbs in terms of likelihood (*likely*, *probably*, *streak*). The amount of hedging also follows the verb level on all three opponents (rank correlation .75 to .92).

**Table 2.** Accuracy percentage per verb level with no hint. Instruct model, Generation , one seed, 43 probes per 100 rounds per verb.

IN = standard harness, RC = harness where model is prompted to give a rule after prediction (of how it was calculated).

| verb level                         | mirror (IN) | mirror (RC) | cycle (IN) | tft (IN) |
|------------------------------------|-------------|-------------|------------|----------|
| high (predict, forecast, infer)    | 45.5        | 50.8        | 97.7       | 39.0     |
| mid (expect, anticipate, estimate) | 37.7        | 41.9        | 94.6       | 35.1     |
| low (guess, suppose, imagine)      | 30.2        | 49.6        | 90.7       | 43.4     |
| order high > mid > low?            | yes         | no          | yes (note) | no       |

Note: on cycle the gap comes only from *suppose* and *imagine*, which as imperatives often answer without the use of reasoning.

The expected order appears cleanly on only one of the four databases (Table 1), And from the reasoning columns of these databases it was deduced that predict opens with the cycle/repeating pattern hypothesis no matter the opponent, while infer does the most cross-stream checking and writes twice as many correctly lagged pairs as any other verb.

**Table 3.** Study 2, infer minus predict. Brackets show on how many of the five opponents infer was higher.

| level | what the hint adds                                                 | accuracy | checks own moves | states a rule |
|-------|--------------------------------------------------------------------|----------|------------------|---------------|
| L0    | nothing (same prompt in both arms)                                 | +0.3     | -0.1 (2)         | -1.0 (1)      |
| L1    | Your task is to {verb} the opponent's next move from the history.  | +1.6     | +3.2 (2)         | +4.1 (4)      |
| LN    | <b>L1</b> + "The opponent is not repeating a fixed sequence"       | +3.1     | +6.9 (5)         | +7.2 (5)      |
| LS    | <b>L1</b> + "... depends on something other than its own moves"    | -1.3     | +10.4 (5)        | +7.6 (5)      |
| L2    | <b>L1</b> + "... depends on what you played"                       | -0.1     | +0.6 (3)         | -0.7 (1)      |
| L3    | <b>L1</b> + "... depends on what you played in the previous round" | +0.2     | -1.3 (0)         | +1.0 (4)      |

...= {the opponent's move}

When the hint rules out the easy answer but does not name the model's column (LN, LS), *infer* made the model check its own moves more and state a rule more, on all five opponents. Over all reasoning measures, 78 of 225 tests were significant, against 6 of 180 (3.3%) on the control levels. But accuracy barely moved: only LN improved (+3.1 points). On mirror, infer gained +5.4 (L1) and +3.4 (LN), not significant. With the hint (L2, L3) the two verbs give the same accuracy and reasoning.

For comparison, one true hint sentence (**L2**) moved accuracy far more than any verb (mirror +47.2, tft +32.7 points), and a false one (**LN**) dropped cycle from 96.2% to 44.6%.

**5. Conclusion.** The verb changes *how* the model reasons, but not *how often it is right*. Prediction accuracy here depends on whether the model looks at the right part of the history (its own previous move w.r.t to opponent), not on the verb used to ask. Better wording alone therefore does not close the gap; the bottleneck is which hypothesis the model tries first and how often it can stick to the correct hypothesis.

## 6. References

Riemer, M., Ashktorab, Z., Bouneffouf, D., Das, P., Liu, M., Weisz, J. D., & Campbell, M. (2025). Position: Theory of Mind Benchmarks are Broken for Large Language Models. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, PMLR 267, pp. 82091–82130.